Get Ready for the Storm

Automating Name Authority Records Update

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What this presentation is/is not

- Automates update of existing name authority records in Millennium
- Works best with SkyRiver
- Series authority records excluded … currently
- Bibliographic File Maintenance … next step

Agenda

- Background
- Goals/Tasks
- Implementation Options
- Workflow (Initial Design)
- Issues & Solutions
- Revised Workflow
- Limitation & What’s Next?
- Demo

Acronyms

- PCC (Program for Cooperative Cataloging)
  - NACO, SACO, CONSER, BIBCO
  - NACO (Name Authority Cooperative Program)
- NAR (Name Authority Record)
- LC/NACO NAF (Name Authority File)
- BFM (Bibliographic File Maintenance)
  - Heading updates in bib records

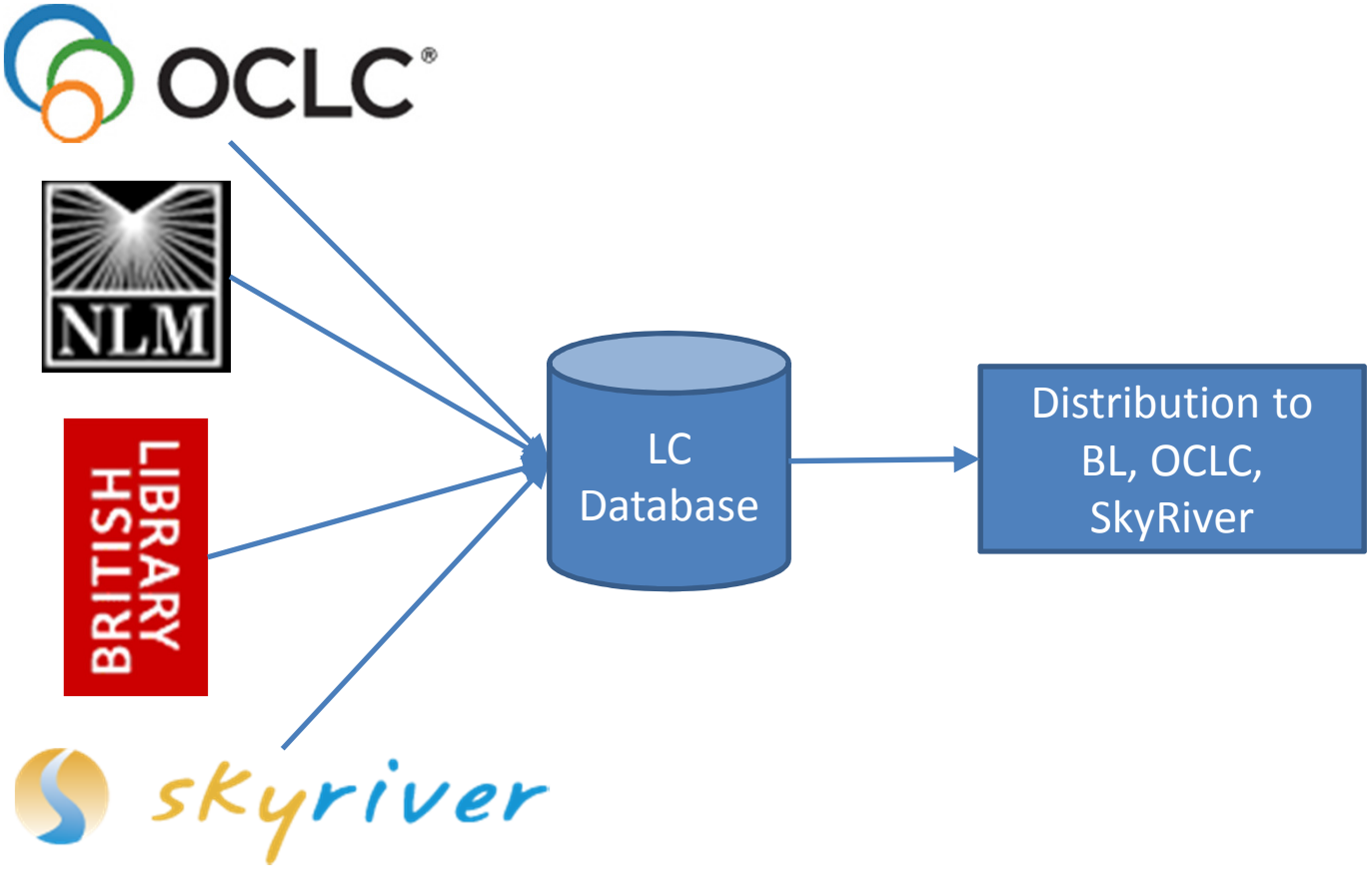
Acronyms (cont'd)

- RDA (Resource Description and Access)
  - Content Standard
- MARCXML (MARC in XML)
- SRU (Search/Retrieval via URL)
  - HTTP request
- XSLT (eXtensible Stylesheet Language Transformation)

LC/NACO NAF

- Dynamic file
- Contributions from over 700 NACO participants
- Updated everyday with new and changed NARs from NACO nodes
  - Full nodes
    - British Library, OCLC, SkyRiver
  - Contribution-only node
    - National Library of Medicine

LC/NACO NAF Maintenance



Authority Control at MSU

- In-house
  - NACO institution
  - Database maintenance
- Post-cataloging Authority Control
  - New Headings Report
    - Download NARs from SkyRiver
  - Updates to NARs not necessary caught
    - 1XX (No item cataloged under changed 1XX → not in new heading report)
    - Elements other than 1XX (e.g. 4XX, 670)

LC/NACO NAF RDA Transition

- PCC Day 1 for RDA NAR: Mar. 31, 2013
- PCC Task Group on AACR2 & RDA Acceptable Heading Categories (Aug 2011)
  - 225,000 NARs with 1XX not usable in RDA bib records
  - 172,000 NARs with 1XX usable in RDA bib records after batch manipulation by software
  - 7,631,00 NARs with 1XX usable in RDA bib records as they are and can be recoded as RDA

LC/NACO NAF RDA Transition (Cont'd)

- Phased reissuance of NARs
  - Phase 1:
    - Scope
      - NARs with characteristics known to be at variance with RDA practice
      - Not candidates for any of the mechanical changes to be made during phase 2
    - Adding a 667 note "THIS 1XX FIELD CANNOT BE USED UNDER RDA UNTIL THIS RECORD HAS BEEN REVIEWED AND/OR UPDATED"
      - Completed Aug. 20, 2012 (436,943 records processed)
  - Phase 2
    - Programmatic changes to 1XX headings that are not acceptable under RDA (e.g., changes to Bible headings, spelling out Dept. and months, etc., abbreviations in the subfield $d for personal names)
    - Scheduled to take place before March 31, 2013

LC/NACO NAF RDA Transition (Cont'd)

- Updates of NARs by NACO institutions
  - Reviewing, upgrading, and recoding Phase 1 records to RDA
  - Upgrading and recoding non-Phase 1 records to RDA
  - Adding any of the 17 new MARC fields (e.g. 046, 372, etc.)
  - Routine NAR maintenance (under both AACR2 & RDA)

Tasks

- To download NARs one-by-one/in bulk
- To detect updates to NARs already in Millennium
- To overlay existing NARs with updated ones
- To automate the above three tasks

Task #1: Download NARs

- Free or already paid-for sources of NARs
  - id.loc.gov
  - LC Permalink Service
  - OCLC LCNAF SRU Service
  - SkyRiver

Task #1: Download NARs(cont'd)

- id.loc.gov (http://id.loc.gov/download/)
  - Pros
    - Service from LC
    - Linked data
    - Bulk download
  - Cons
    - Not very up-to-date (Feb/Mar 2012)
    - Only available in MADS and SKOS

Task #1: Download NARs (cont'd)

- LCCN Permalink Service (http://lccn.loc.gov/lccnperm-faq.html)
  - Pros
    - Service from LC
    - SRU-based service (HTTP request)
    - Available in MARCXML & MADS
    - Retrievable by LCCN (e.g. http://lccn.loc.gov/n79018774/marcxml)
  - Cons
    - One-by-one search & retrieval
    - Discourages software from submitting over 10 requests/minute
      - Terminate programs that require more than 24 hours to complete
    - No crawling by robots
      - Block IP addresses that fail to honor the LCCN Permalink's robots.txt file

Task #1: Download NARs (cont'd)

- OCLC LCNAF SRU Service (http://alcme.oclc.org/srw/search/lcnaf)
  - Pros
    - Multiple indexes (LCCN, names, dates, etc.)
    - Available in multiple schema including MARCXML
    - One-by-one or bulk download*
    - SRU-based service (HTTP request)
  - Cons
    - Updated every Monday night
    - OAI-PMH service is not available though there is an index for OAI identifier
    - Bulk download – by search term (e.g. after certain date)

Task #1: Download NARs (cont'd)

- SkyRiver
  - Pros
    - Left-handed and keyword search available
    - Keyword search indexes almost all variable fields
  - Cons
    - One-by-one search and retrieval
    - Slower than other SRU-based services
    - LCCNs in 010$z searched as keyword
    - Available only in MARC

Task #1: Download NARs (cont'd)

- Source picked: OCLC SRU Service
  - Search LCCNs one-by-one
    - 10 records/sec. retrieved by program
  - Download XML files into one folder (files named by LCCN)

Task #2: NAR Update Detection

- To compare NARs from Millennium and NARs from LC/NACO NAF
  - MARC 005 (timestamp)

MARC Leader			#####cz 22#####n 45 0
MARC	001		oca00278908
MARC	003		OCoLC
MARC	005		20061128071359.0
MARC	008		790611n| acannaabn |a aaa
LC ARN	010		n 79045521
MARC	040		DLC|beng|cDLC|dDLC|dDLC-R|dOCoLC|dFMU|dDLC|dWaU
MARC	053	0	PS3525.I5156|5MiEM
NAME AUTHR	100	1	Miller, Arthur,|d1915-2005
NAME S FRM	400	1	Miler, Artur,|d1915-2005
NOTE	670		His Situation normal, 1944.

  - If timestamp more current on the NAR from NAF → Overlay the NAR in Millennium

Task #2: NAR Update Detection (Cont'd)

- XSLT (eXtensible Stylesheet Language Transformation)
  - Within the family of XML
  - Current version: 2.0
  - Unicode compliant
  - Multiple inputs, multiple outputs
  - "Transformation" means:
    - Manipulation of documents by creating a new document based on the original document
    - Output can be in XML, HTML, XHTML, text
  - Data-driven execution
    - Codes executed when a certain piece of data is encountered → unexpected outcomes

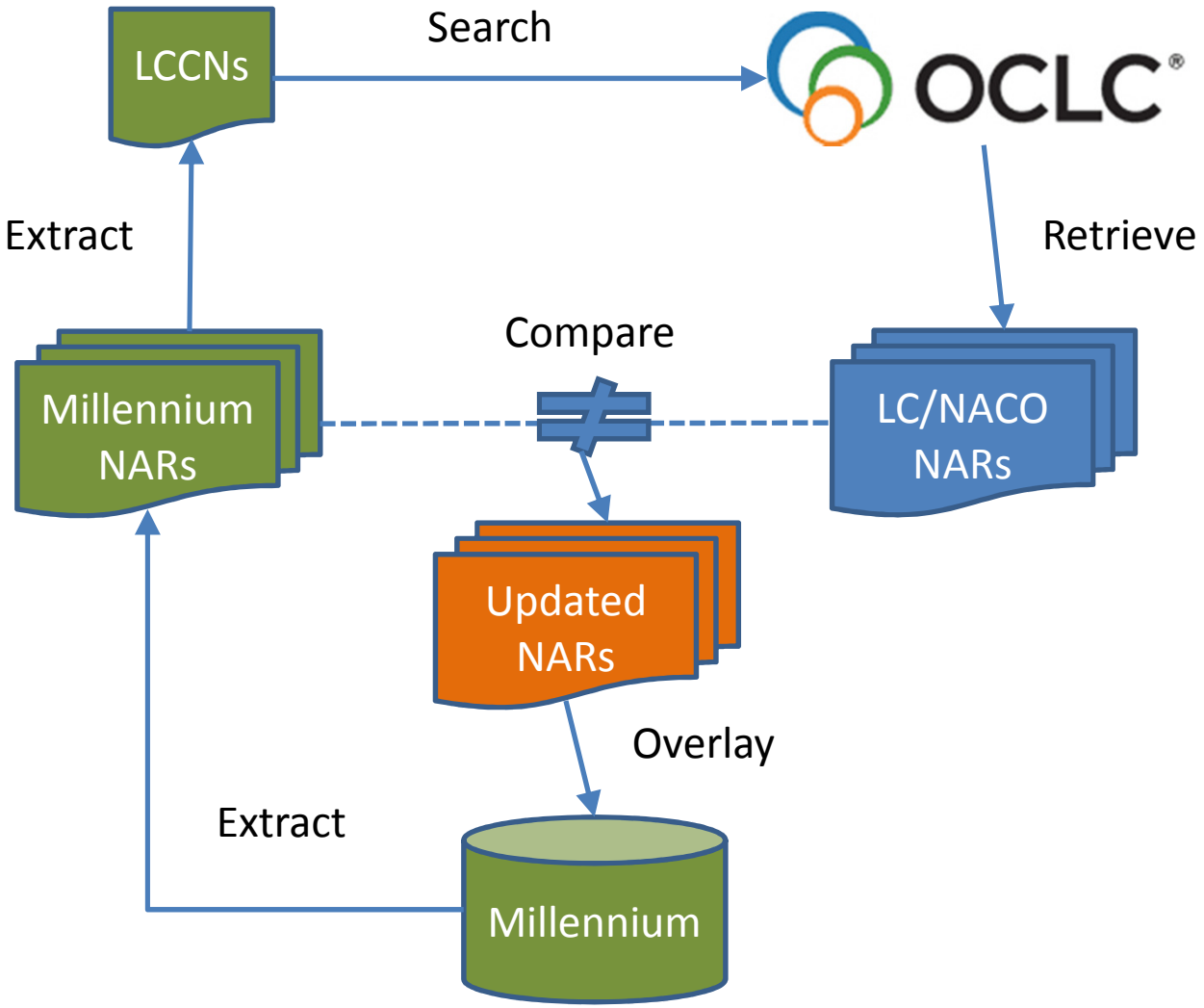
Task #3: Export/Overlay of NARs

- MarcEdit
  - Export MARC Records (Bib & Auth)
  - Through TCP/IP (Host address, Port, .mrc file)
  - Same as export from OCLC Connexion or SkyRiver
  - One-by-one (though .mrc file can contain multiple NARs)
- Data Exchange (Millennium)
  - Record load
  - Not authorized to use

Task #4: Automation

- To link up various steps in the workflow
- AutoIt (http://www.autoitscript.com/site/autoit/)
  - Freeware
  - Script editor
  - Designed for automating the Windows GUI and general scripting
    - Simulated keystrokes, mouse movement and window/control manipulation
    - Simple text manipulation (supports regular expression), read/write text file
    - Send HTTP request
    - Open/run/close applications

Workflow


LCCNs
Search
OCLC
Extract
Retrieve
Compare
Millennium NARs
LC/NACO NARs
Updated NARs
Overlay
Extract
Millennium

Data Integrity Issue #1

- No Millennium ARN in extracted NARs
  - Needed for 949 overlay command
  - Solution
    - Extract “LCCN/035” & “ARN” pair through Create Lists
    - Merge ARN into extracted NARs (907$a) by XSLT/MarcEdit

Data Integrity Issue #2

- NARs without 010
  - 010 contains LCCN
  - Some LCCNs transposed into 035
    - Original prefix (n, no, nb, nr) removed
    - Prepended with prefix (OCoLC)
    - Possibly done during system migration
  - Solution
    1. Search string in 035 (excl. prefix) as keyword in SkyRiver
    2. Retrieve complete LCCN from matched record
    3. Search retrieved LCCN against OCLC Service and download the record

Data Integrity Issue #3

- Existing NARs without 005
  - No timestamp
    - Bring in the new NAR whenever the old NAR lacks 005, but …

Data Integrity Issue #4

- Same LCCN, Different Entities/Identities
  - Shift of identities in NAR through time
  - When comparing NARs, check the existence of:
    - Old heading in the new NAR
    - 1st 670$a of old NAR in the new one

Data Integrity Issue #5

- Local data in NAR
  - Local call no. (e.g. 050, 090, 053$5)
  - Institution code & initials (shared catalog)
  - Copy local data into new NAR before overlay

Search and Retrieval Issue #1

- "Blank" XML File from OCLC SRU Service

```
<?xml version="1.0" ?>
<?xml-stylesheet type="text/xsl" href="/SRW/searchRetrieveResponse.xsl"?>
<searchRetrieveResponse xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://www.loc.gov/zing/srw/ http://www.loc.gov/standards/sru/sru1-1archive/
    xmlns="http://www.loc.gov/zing/srw/">
    <version xsi:type="xsd:string">1.1</version>
    <numberOfRecords xsi:type="xsd:nonNegativeInteger">0</numberOfRecords>
    <echoedSearchRetrieveRequest xsi:type="ns1:echoedSearchRetrieveRequestType"
        xmlns:ns1="http://www.loc.gov/zing/srw/">
        <version xsi:type="xsd:string">1.1</version>
        <query xsi:type="xsd:string">local.LCCN = &amp;quot;n80066947&amp;quot;</query>
        <xQuery>
            <ns2:searchClause xsi:type="ns2:searchClauseType"
                xmlns:ns2="http://www.loc.gov/zing/cql/xcql/">
                <ns2:index xsi:type="xsd:string">local.LCCN</ns2:index>
                <ns2:relation xsi:type="ns2:relationType">
                    <ns2:value xsi:type="xsd:string">=</ns2:value>
                </ns2:relation>
                <ns2:term xsi:type="xsd:string">n80066947</ns2:term>
            </ns2:searchClause>
        </xQuery>
        <startRecord xsi:type="xsd:positiveInteger">1</startRecord>
        <maximumRecords xsi:type="xsd:nonNegativeInteger">10</maximumRecords>
        <recordPacking xsi:type="xsd:string">xml</recordPacking>
        <recordSchema xsi:type="xsd:string">info:srw/schema/1/marcxml-v1.1</recordSchema>
        <resultSetTTL xsi:type="xsd:nonNegativeInteger">300</resultSetTTL>
        <sortKeys xsi:type="xsd:string"/>
    </echoedSearchRetrieveRequest>
```

Search and Retrieval Issue #1 (Cont'd)

- No hit for some LCCNs
  - XML file size: < 2KB
  - LCCNs in places other than 010$a → Not indexed
    - Cancelled LCCNs (010$z)
    - LCCNs in 667
- Solution
  1. Compile a list of LCCNs with file size < 2KB
  2. Search LCCNs in SkyRiver by Keyword
  3. Get new LCCNs from 010$a
  4. Search OCLC SRU Service using new LCCNs
  - But …

Search and Retrieval Issue #2

- Keyword search in SkyRiver returns multiple hits
  - Undifferentiated & related NARs

010			n 2004021255
040			DLC **|b** eng **|c** DLC
100	1		Williams, Allan, **|d** 1930-
667			Formerly on undifferentiated name record: n 50017337
670			His The man who gave the Beatles away, 1975
670			BL Web OPAC, Mar. 30, 2004 **|b** (Williams, Allan, 1930-)
670			BL auth. fiche, Sept. 2000 **|b** (Williams, Allan, 1930-; Allan Richard; b. Feb. 21)

Person broken out from undifferentiated NAR

010			n 92032058 **|z** n 50017337
040			DLC **|c** DLC **|d** DLC
100	1		Williams, Allan N.
670			Bachmann, W.S. A listing of environmental permits ... 1991: **|b** t.p. (Allan N. Williams)
670			His Flood management in Connecticut, 1980: **|b** t.p. (Allan Williams, Natural Resources Center, Dept. of Environment Protection)

Original undifferentiated NAR cancelled

  - Write LCCNs with multiple hits to a log file for manual review

Search and Retrieval Issue #2 (Cont'd)

- Keyword search in SkyRiver returns multiple hits
  - Same numeral part of LCCN with different prefixes

010			nr 89006300
040			NjP **|c** NjP
100	1		Pagel, Hans-Joachim
670			Weissenborn, T. Alchimie, 1987: **|b** t.p. (Hans-Joachim Pagel)

NAR contributed via RLIN

010			no 89006300
040			CLU **|c** CLU
100	1		Arliss, George, **|d** 1868-1946
400	1		Andrews, George Augustus, **|d** 1868-1946
670			The House of Rothschild [MP] 1934: **|b** credits (players, George Arliss)
670			Halliwell's Filmgoer's comp., 1985 **|b** (Arliss, George; b. 1868 d. 1946; AKA George Augustus Andrews)

NAR contributed via OCLC

  - Write LCCNs with multiple hits to a log file for manual review

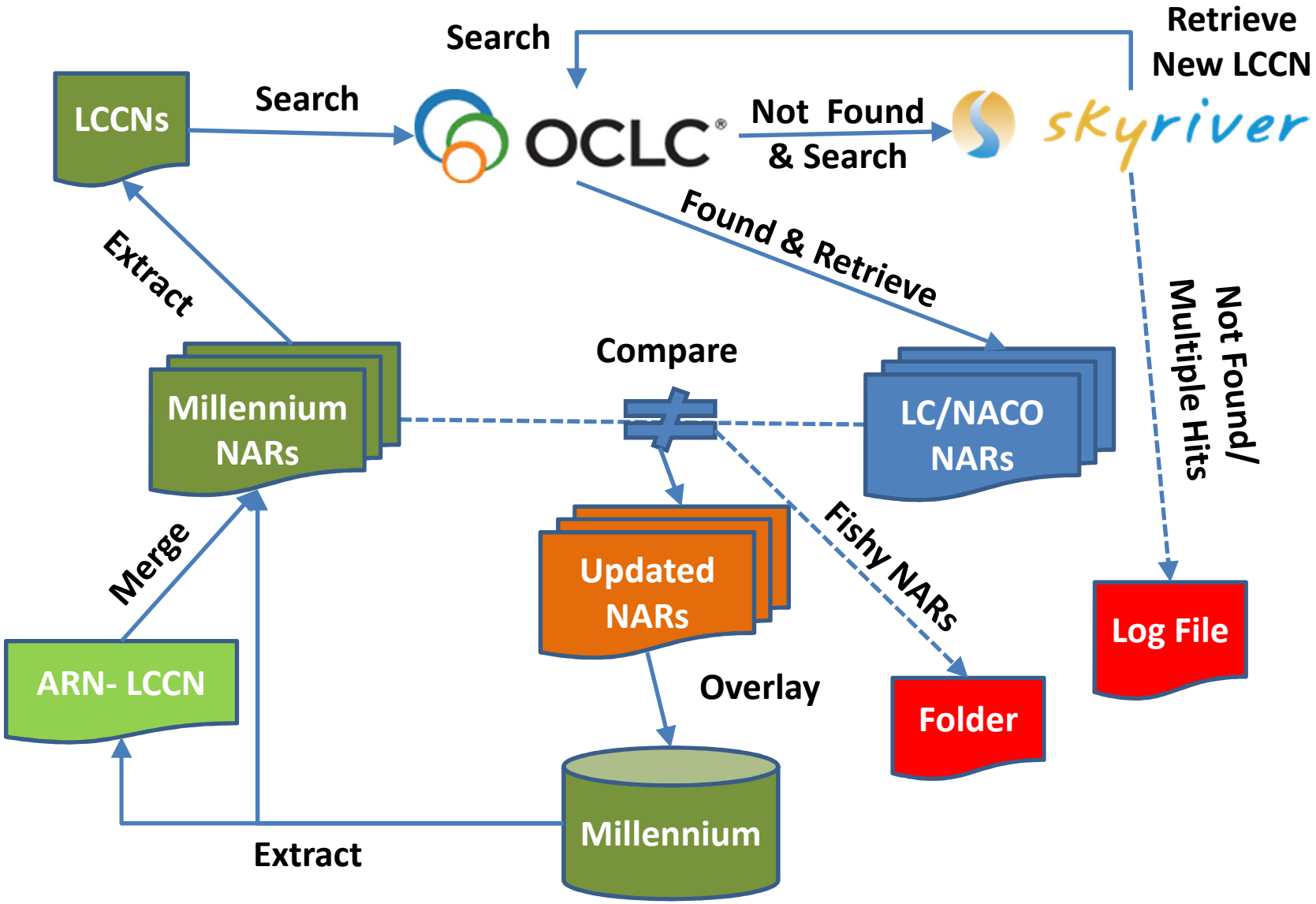
Search and Retrieval Issue #2 (Cont'd)

- Keyword search in SkyRiver returns no hit
  - The LCCN in question no longer exists in NAF
    - NAR containing cancelled LCCN was cancelled again
      - Loss of 010$z
    - Write no-hit LCCNs into log file for manual review

Search and Retrieval Issue #2 (Cont'd)

- Keyword search in SkyRiver returns no hit
  - False negative
    - Space between prefix and number removed
    - Hyphen within number removed (e.g. n 85-342238 → n 85342238)
      - Search normalized LCCNs
    - Delay in returning result for a search due to slow or unstable Internet connection speed
      - Set a longer wait time before trying to copy new LCCN
      - Run keyword search in SkyRiver in loop until
        - » Number of entries in log file equals to immediate preceding round, or
        - » File size of the no-hit log file equals zero

Revised Workflow



Limitations

- Identities broken out from undifferentiated NARs can’t be detected
  - Partially taken care of by “New Headings Report”
- Slow export using MarcEdit
- Millennium NARs with no LCCNs
  - OCLC ARN not indexed by OCLC SRU Service
  - Search by text-string unreliable (e.g. multiple hits)

What's Next?

- BFM following NAR update
  - Compare 1XXs in Millennium NARs and LC/NACO NARs
  - Compile a conversion table of old and new 1XXs
  - Use Millennium Global Update to complete heading changes
- Update AutoIt script for Sierra
- Series Authority Records

Demo & Q&A

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